

		when the new temperature is 97, the new r.m.s. speed – 3v
15	D	Option A: The kinetic energy of the oscillator is proportional to square of the frequency of its motion. Incorrect. Option B: The potential energy of the oscillator is maximum when the oscillator is momentarily at rest. Incorrect. Option C: The kinetic energy of the oscillator is maximum when the oscillator is at the equilibrium position. Incorrect. Option D: When the kinetic energy of the oscillator is equal to its potential energy, the oscillator is neither at the rest position nor at the maximum displacement positions. Correct.

Qn	Ans	Solution
16	B	Using $y = x \cdot a$